

Low-cost transport data collection: Making use of pre-existing data with computer vision

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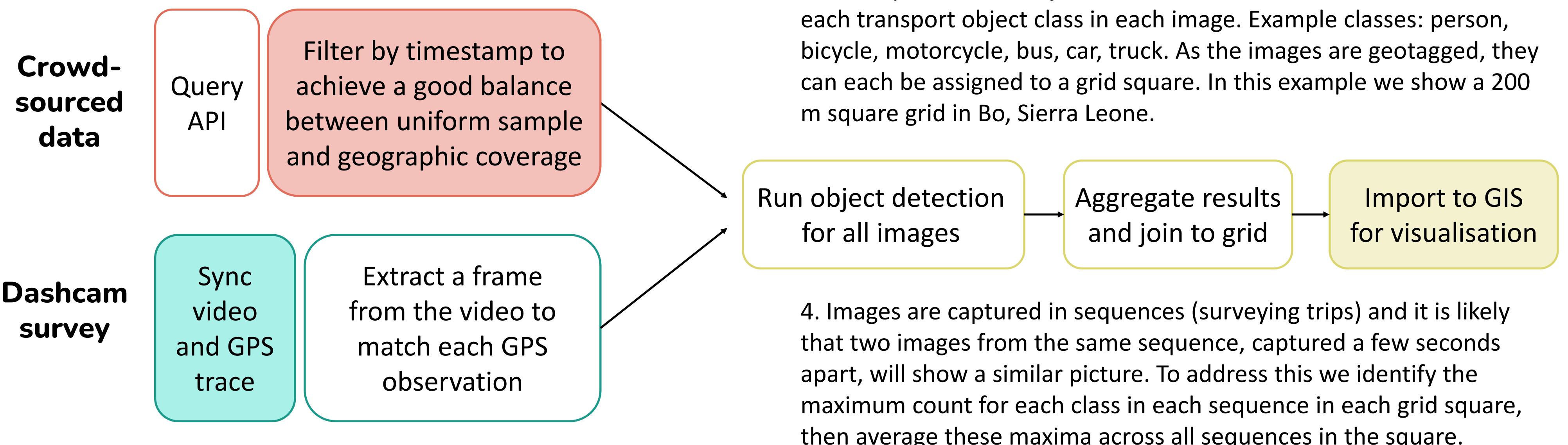
Cities in low and middle-income countries (LMICs) often have inadequate transport systems which fail to enable safe, affordable and efficient travel. Many local government bodies responsible for transport lack the resources to collect data, which affects their ability to effectively regulate the transport system and develop improvement schemes.

To address this issue we have developed a software tool to map the spatial distribution and relative prevalence of transport modes across a city (see maps below).

We intend the tool to be used to produce high-level information across a wide geographic area at a very low cost, not to replace classified traffic surveys or to inform detailed engineering design. The primary use case is expected to be in transport strategy development, where wide coverage is more useful than fine levels of detail, but it could also be used to help target transport interventions or familiarise staff with a new city.

Data collection tool workflow:

1. Obtain a set of geotagged street-level images. The tool is built around freely available crowdsourced data (Mapillary.com). Pre-existing data has variable quality and availability, however, so we also tested the use of bespoke dashcam video captured with an accompanying GPS trace. Although much more expensive, this allows control over critical aspects such as time of day or day of week.



2. In the interests of creating a sample showing similar travel demand, we filter the set by timestamp so that all images were captured at a similar time of day and day of the week. There is a trade-off between sample uniformity and geographic coverage, which will likely vary on a project-by-project basis.

3. A computer vision object detector is used to count the number of each transport object class in each image. Example classes: person, bicycle, motorcycle, bus, car, truck. As the images are geotagged, they can each be assigned to a grid square. In this example we show a 200 m square grid in Bo, Sierra Leone.

4. Images are captured in sequences (surveying trips) and it is likely that two images from the same sequence, captured a few seconds apart, will show a similar picture. To address this we identify the maximum count for each class in each sequence in each grid square, then average these maxima across all sequences in the square.

5. Results are presented on a relative scale as the nature of imagery from a moving vehicle means we cannot accurately determine the absolute number of each object class in each area. We can, however, use these outputs to identify areas of interest for classified traffic counts, if desired.

Key areas for development:

1. Improve detection reliability (particularly three wheelers, vans and heavy trucks, but also far-away objects) → train custom CV model
2. Testing and validation to increase confidence in the tool's outputs → compare outputs against area-wide traffic count data

